

2.2: Graphs of Linear Functions

Definition. (Parallel and Perpendicular lines)

Let L_1 and L_2 be lines with slopes m_1 and m_2 respectively. If L_1 and L_2 are *parallel*, then

$$m_1 = m_2.$$

If L_1 and L_2 are *perpendicular*, then

$$m_1 = -\frac{1}{m_2}.$$

Example.

Find the line parallel to $y = \frac{3}{2}x + 1$ that passes through the point $(-4, 10)$.

$$m = \frac{3}{2}$$

$$y - 10 = \frac{3}{2}(x - (-4))$$

$$y = \frac{3}{2}x + 6 + 10$$

$$y = \frac{3}{2}x + 16$$

Find the line perpendicular to $y = \frac{3}{2}x + 1$ that passes through the point $(-3, 4)$.

$$m = -\frac{2}{3}$$

$$y - 4 = -\frac{2}{3}(x - (-3))$$

$$y = -\frac{2}{3}x - 2 + 4$$

$$y = -\frac{2}{3}x + 2$$

Definition.

The point(s) where a graph intersects the axes are called intercepts.

The **x -intercepts** are the points where a function intersects the x -axis.

The **y -intercept** is the point where a function intersects the y -axis.

- To solve for the y -intercept:
 - Set $x = 0$,
 - Solve for y .
- To solve for the x -intercept:
 - Set $y = 0$,
 - Solve for x .

Example. Find the x and y intercepts of the following:

$$y = \frac{4}{5}x - 8$$

$$\begin{array}{l} \text{y-int (x=0)} \\ y = \frac{4}{5}(0) - 8 = -8 \Rightarrow \boxed{(0, -8)} \end{array}$$

$$\begin{array}{l} \text{x-int (y=0)} \\ 0 = \frac{4}{5}x - 8 \\ 8 = \frac{4}{5}x \\ 10 = x \Rightarrow \boxed{(10, 0)} \end{array}$$

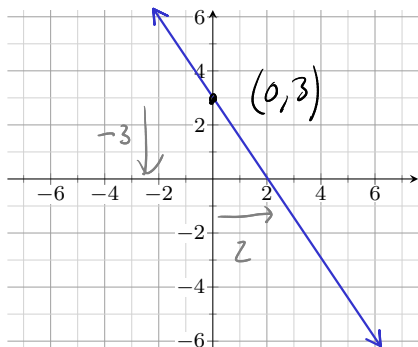
$$3x + 2y = 18$$

$$\begin{array}{l} \text{y-int (x=0)} \\ 3(0) + 2y = 18 \\ 2y = 18 \Rightarrow \boxed{(0, 9)} \\ y = 9 \end{array}$$

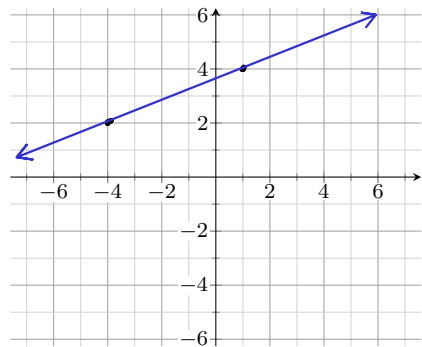
$$\begin{array}{l} \text{x-int (y=0)} \\ 3x + 2(0) = 18 \\ 3x = 18 \Rightarrow \boxed{(6, 0)} \\ x = 6 \end{array}$$

Example. Graph the following:

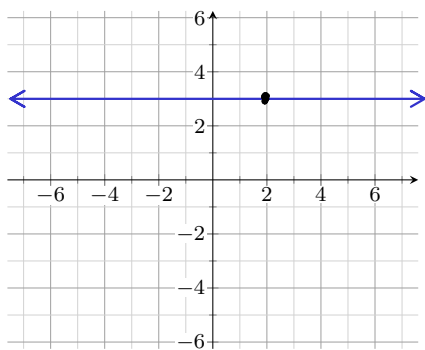
$$y = -\frac{3}{2}x + 3$$



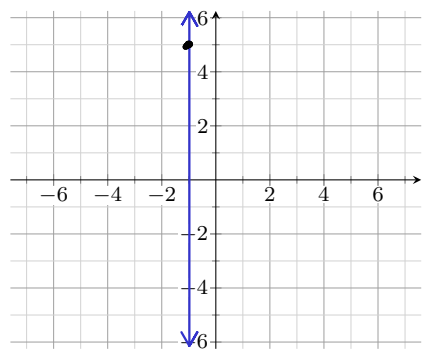
The line through $(-4, 2)$ and $(1, 4)$



The horizontal line through $(2, 3)$

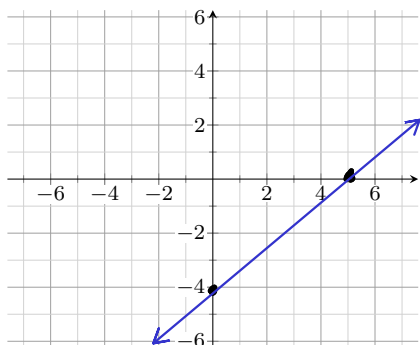


The vertical line through $(-1, 5)$

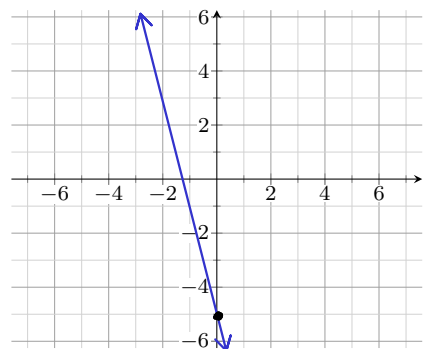


$(5, 0)$
 $(0, -4)$

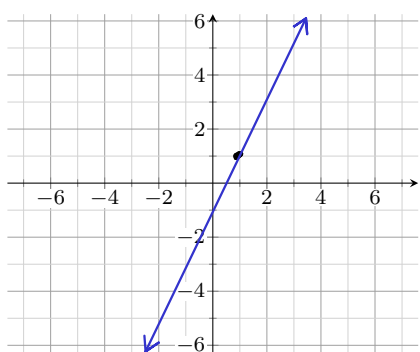
$$8x - 10y = 40$$



$$y = -4x - 5$$

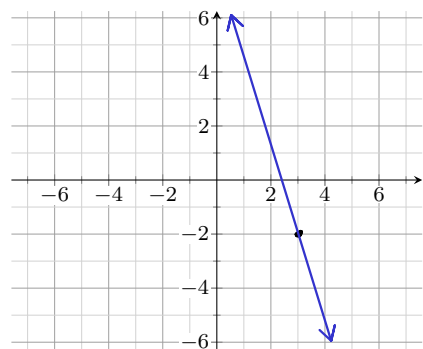


Through $(1, 1)$, parallel to $y = 2x - 4$



$m = 2$

Through $(3, -2)$, perp. to $y = \frac{1}{3}x - 1$



$m = -3$